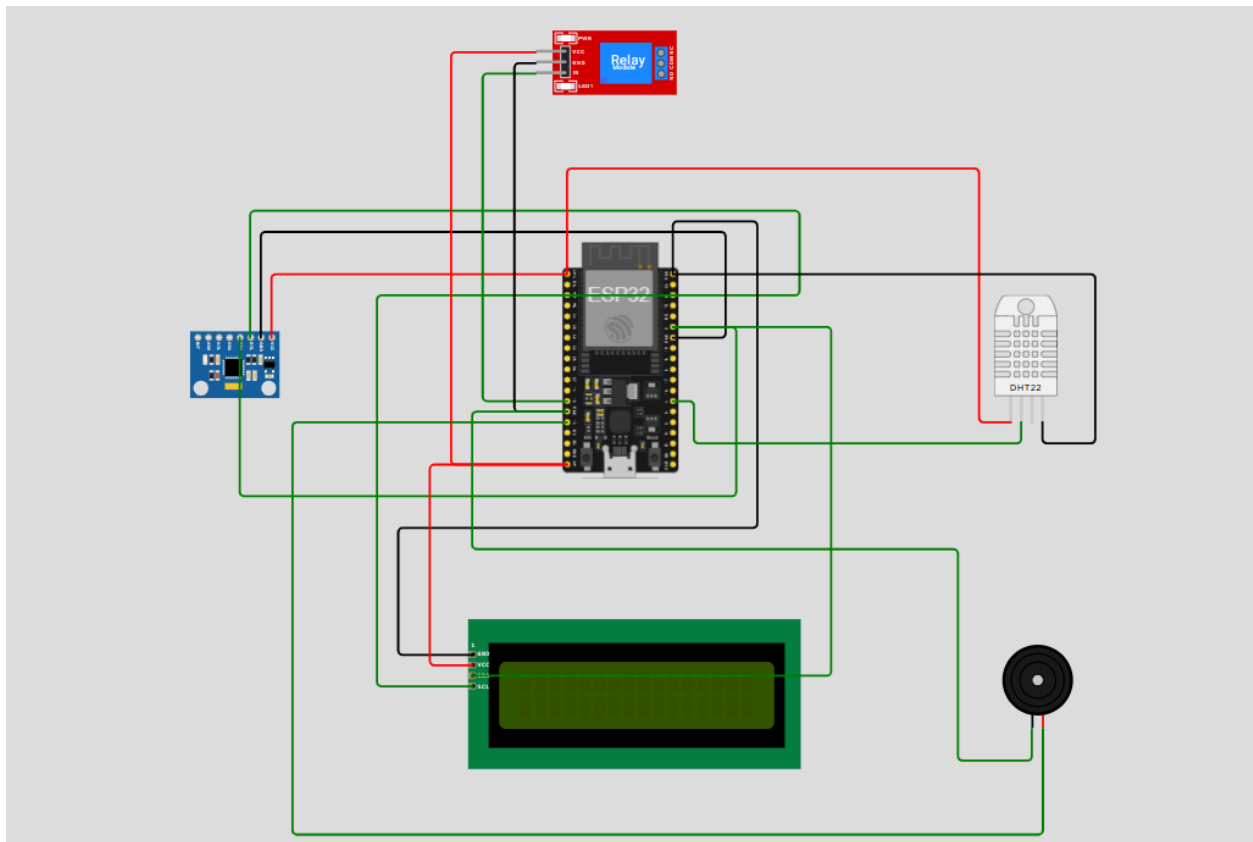


CIRCUIT DIAGRAM:



Power Distribution

Component	Supply Pin	ESP32 Connection
All Components	GND	Common Ground Rail
I2C LCD Display	VCC	5V / VIN
Relay Module	VCC	5V / VIN

MPU6050 Accelerometer	VCC	3.3V
DHT22 Temperature Sensor	VCC	3.3V

Communication & Signal Pins

1. Shared I2C Bus

- **SDA (Data):** Connect both MPU6050 SDA and LCD SDA to GPIO 21
- **SCL (Clock):** Connect both MPU6050 SCL and LCD SCL to GPIO 22

2. Dedicated Digital I/O

- **DHT22 Sensor:** Connect Data pin to GPIO 4
- **Relay Module:** Connect IN / Signal pin to GPIO 12
- **Piezo Buzzer:** Connect Positive (+) pin to GPIO 13
- **Status LED:** Connect Anode (+) leg to GPIO 14 (*Cathode connects through a \$220\,\Omega\$ resistor to Ground*)

Master Pinout Summary Table

Pin	Connected Device	Mode	Type
GPIO 4	DHT22 Sensor	INPUT	Digital Pulse
GPIO 12	Relay Module	OUTPUT	Digital (HIGH/LOW)

GPIO 13	Piezo Buzzer	OUTPUT	PWM Tone
GPIO 14	Status LED	OUTPUT	Digital (HIGH/LOW)
GPIO 21	MPU6050 & LCD	BIDIRECTIONAL	I2C Data (SDA)
GPIO 22	MPU6050 & LCD	OUTPUT	I2C Clock (SCL)